

LIGHTWEIGHT SAFE HELICOPTER

LSH 00 — COMPREHENSIVE ENGINEERING BASELINE

Generation 1 | Revision 3.3 — Multi-Layer Capsule Protection Architecture
Engineering-Strengthened Preliminary Controlled System Baseline

Document ID	LSH 00
System Name	Lightweight Safe Helicopter
Variant focus	Generation 1 platform family — common survival capsule concept
Revision	3.3 — Refined multi-layer capsule protection & planning mass update
Status	[PROPOSED] Preliminary Controlled Baseline
Founder	Amirbahador Atai
Language	English
Certification status	Not a certification approval; no flight or performance claim is demonstrated
Primary use	Root system baseline for requirements, architecture, analysis, V&V,; certification and licensing documentation

Engineering integrity statement: LSH 00 is a system-level preliminary baseline. It deliberately does not present unverified aircraft performance, emergency-separation capability, crashworthiness, parachute performance, airbag performance, seat-damper performance, reliability or certification compliance as facts. Any item that requires analysis, test, inspection or certification evidence remains explicitly open. All mass, power, timing, G-load attenuation and performance figures are planning values only and carry no demonstrated status.

Revision 3.3 strengthening focus: Capsule primary structure defined as a one-piece CFRP plus-shaped (+) roll-cage with substantial cross-section members; distributed 12-airbag architecture; thick energy-absorbing seats (≥ 50 cm) with dual mechanical dampers between each seat and the roll-cage; minimum four-point shock-isolated capsule-to-airframe attachments; progressive mechanical floor attenuation; multi-layer multi-nozzle fire suppression for residual failed-separation cases; planning MTOW updated to 20 000 kg; rotor and power retained strictly as study ranges for a planning thrust-to-weight objective of 1.6. Partial and asymmetric separation remain first-class design cases. No quantitative survival probability at extreme impact loads is claimed.

1. PURPOSE, SCOPE AND BASELINE BOUNDARY

This document defines the system-level baseline for a twin-engine helicopter family that incorporates an occupied survival capsule and a controlled emergency separation / jettison function. It provides the root structure from which a large engineering document tree (hundreds of controlled documents) can be decomposed without silently converting assumptions into requirements.

The baseline defines requirements, targets, assumptions, calculated quantities, verification gates, interfaces, hazards and configuration-control rules. It also establishes a common Safety Core (capsule structure, attachments, attenuation, airbags, seats, fire suppression and separation logic) that is intended to remain dimensionally similar across family variants so that the same protection architecture can be reused.

Boundary: LSH 00 is not a detailed design, production drawing package, certification approval, flight manual or demonstrated performance report. No statement in this document constitutes a claim of feasibility, airworthiness or operational safety.

2. SYSTEM DESCRIPTION

The baseline aircraft is a conventional twin-turboshaft, single-main-rotor helicopter with a tail rotor and short fixed mechanical landing supports. Nominal occupancy for the baseline variant is seven persons (two flight crew plus five additional occupants). Family variants may increase seating while keeping the capsule external envelope approximately common.

A dedicated survival capsule forms the final occupied landing boundary after any successful emergency separation. The emergency architecture is intended to reduce the mass and fire hazard carried into the capsule's post-separation landing phase. Separation shall receive survival credit only after the relevant verification evidence exists. Normal flight, emergency separation, parachute descent and ground impact are treated as coupled system states, not as independent demonstrations.

3. STATUS AND CONFIGURATION LEGEND

Tag	Meaning	Use
[REQ]	Allocated requirement	Baseline obligation unless changed by formal configuration control
[TARGET]	Design target	Objective subject to engineering closure; not a demonstrated value
[ASSUMP]	Engineering assumption	Temporary analytical basis with identified owner and closure plan
[CALC]	Calculated quantity	Valid only for the stated inputs, equations and configuration
[OPEN]	Open verification / analysis	No compliance or performance claim permitted
[TBD]	Not yet defined	Must be resolved by a controlled downstream document

Tag	Meaning	Use
[DEMO]	Demonstrated	Only after objective evidence is formally accepted

4. TOP-LEVEL SYSTEM REQUIREMENTS (SELECTED)

ID	Requirement / Target	Status
LSH-REQ-SYS-001	The aircraft shall provide seating and restraint provisions for the defined occupancy of each variant.	[REQ]
LSH-REQ-SYS-002	Generation 1 architecture shall use two engines and one main rotor with a tail rotor.	[REQ]
LSH-REQ-SYS-003	The occupied survival capsule shall constitute the defined final landing boundary.	[REQ]
LSH-REQ-CAP-001	The capsule shall include occupant protection, impact attenuation, recovery and post-event survival provisions.	[REQ]
LSH-REQ-JET-001	Emergency separation shall have at least one automatic initiation path and an independent manual crew initiation path; any backup automatic path shall be independently assessed.	[REQ]
LSH-REQ-JET-002	Unintended separation / jettison shall be treated as a catastrophic hazard and shall receive a quantitative safety objective.	[REQ]
LSH-REQ-JET-003	No component shall be designated as universally jettisonable until structural, dynamic, fire, control and collision analyses show that release is acceptable for the applicable envelope.	[REQ]
LSH-TGT-MASS-001	20 000 kg is the Generation 1 planning MTOW for system studies only.	[TARGET]
LSH-TGT-CAP-001	Capsule external envelope shall remain approximately common across family variants to preserve Safety Core commonality.	[TARGET]
LSH-TGT-TW-001	Planning thrust-to-weight objective of 1.6 (hover) is retained as a study target only; power and rotor size remain open.	[TARGET]
LSH-TGT-JET-001	≤ 6 s from validated emergency command to verified capsule-clear state is a development objective only.	[TARGET]
LSH-TGT-AIRBAG-001	Distributed multi-airbag architecture with a planning count of 12 airbags is the initial study baseline.	[TARGET]
LSH-TGT-ATT-001	Minimum four primary capsule-to-airframe attachment points with controlled shock isolation.	[TARGET]

5. MASS PROPERTIES AND CONFIGURATION STATES

The 20 000 kg value is an accounting baseline, not a verified aircraft mass. The mass model must close for all safety-critical configurations, including normal flight, fuel extremes, emergency separation, capsule-only descent, asymmetric release and failed-release cases. Capsule mass shall be decomposed bottom-up before it becomes a controlled design requirement. Mass growth allowance is a program reserve, not evidence of demonstrated margin.

Item	Planning mass	Status
Complete survival capsule (structure + protection systems + seats + allocation)	5 000 – 5 500 kg class	[TARGET]
Main airframe excluding capsule	study allocation	[ASSUMP]
Main rotor + hub + controls	study allocation	[ASSUMP]
Tail rotor + drive	study allocation	[ASSUMP]
Engines + gearboxes + mounts	study allocation	[ASSUMP]
Fuel system + usable fuel	study allocation	[ASSUMP]
Rapid-separation hardware + independent hardware	study allocation	[ASSUMP]
Avionics / electrical / hydraulic / misc.	study allocation	[ASSUMP]
Mass growth allowance	program reserve	[TARGET]
ACCOUNTING TOTAL (planning)	20 000 kg	[CALC]

6. GEOMETRY, ROTOR LOADING AND PROPULSION STUDY RANGES

Rotor and power figures are planning study ranges only. They are derived from the 20 000 kg planning mass and a thrust-to-weight study objective of 1.6. They do not constitute evidence of feasibility. Disk loading is reported as conventional weight / disk area. A separate thrust margin or thrust-to-weight quantity must never be labelled “disk loading”.

Parameter	Planning value	Status
Capsule external form intent	Approximately box / near-square planform for structural commonality and internal volume efficiency; exact ratios TBD by structural and human-factors closure.	[TARGET]
Capsule external envelope (commonality)	approximately 5.2 × 2.5 × 2.0 m class	[TARGET]
Main rotor diameter (study range)	16.5 – 17.5 m	[TARGET]
Corresponding disk area	approx. 214 – 241 m ²	[CALC]

Parameter	Planning value	Status
Conventional disk loading at 20 000 kg	approx. 83 – 94 kg/m ²	[CALC]
Tail rotor diameter	to be closed with main-rotor and anti-torque analysis	[OPEN]
Installed power	open — requires validated power-required model; preliminary momentum estimates for T/W = 1.6 indicate very high demand	[OPEN]
Engine count / type	two turboshaft (architectural requirement)	[REQ]
Fuel capacity	to be closed with mission model, reserves and hot/high conditions	[OPEN]

Note: A planning thrust-to-weight target of 1.6 at 20 000 kg produces high installed-power estimates. This remains an open feasibility item. Power matching, rotor figure of merit, transmission limits, OEI and hot/high performance shall be closed by dedicated analysis before any engine selection is locked.

7. SURVIVAL CAPSULE ARCHITECTURE (MULTI-LAYER PROTECTION)

The capsule is the final occupied landing boundary. The protection architecture is deliberately layered so that residual survival capability remains even when aerial separation is incomplete or parachutes do not fully function. All elements below are planning definitions pending structural, dynamic and occupant-protection analysis.

7.1 Primary structure — CFRP plus-shaped (+) roll-cage

One-piece (or controlled multi-piece with permanent joints) CFRP primary structure arranged as a plus-shaped (+) roll-cage. Substantial cross-section members extend to the four principal directions. Additional circumferential and longitudinal reinforcing members create a continuous strong cage so that the external surface of the capsule remains structurally coherent under impact. Metallic / titanium nodes are used at critical joints and attachment points. The primary cross-member depth is intended to be in the order of 1 m class (exact geometry TBD by loads and packaging analysis). Status: [TARGET/REQ].

7.2 Entry / exit doors and hatches

Recommended baseline: two side doors (port and starboard) for normal and emergency egress, plus one optional overhead or rear emergency hatch if required by the emergency-egress analysis. A floor hatch is generally avoided because it conflicts with the primary landing-attenuation structure and sealing. A pure four-door arrangement (floor + ceiling + two sides) is not adopted as the baseline because of structural discontinuity, sealing and impact-load-path concerns. Exact count, size and location remain [REQ/TBD] pending human-factors and ditching / crash egress assessment.

7.3 Seats and seat-to-cage isolation

Energy-absorbing seats with a minimum effective thickness of 50 cm, broad sofa / mattress-like support surface and full occupant restraint system. Each seat is mounted to the CFRP roll-cage through dual mechanical dampers / shock absorbers that provide an additional progressive energy-absorption stage between the occupant and the primary structure. Load-stroke law, stroke length and bottoming behaviour are open and must be closed with occupant injury criteria. Status: [TARGET/REQ].

7.4 Floor mechanical attenuation

Progressive mechanical energy-absorption system at the capsule floor (short, thick stroke elements with defined load-stroke characteristics). Failed-leg and asymmetric landing cases are mandatory design cases. Interaction with the airbag system and seat dampers must be analysed as a combined attenuation chain. Status: [REQ/TBD].

7.5 Distributed airbag system

Planning baseline: 12 airbags distributed to cover the floor, side walls and longitudinal (fore/aft) directions. Dual activation logic is required: (1) after successful parachute deployment and near-ground sensing, and (2) residual hard-impact activation when separation is incomplete or parachutes have not provided the expected descent rate. Coverage, inflation time, pressure, reliability and interaction with seats and structure remain open. Status: [TARGET].

7.6 Capsule-to-airframe attachments

Minimum four primary load-path attachment points providing controlled shock isolation / progressive energy absorption in normal flight and a clean release path in emergency. Each attachment must be analysed for normal-flight loads, emergency release, failed-release, unintended-release and residual structural integrity after partial separation. Status: [TARGET/REQ].

7.7 Multi-layer fire detection and suppression

Multi-layer, multi-nozzle fire detection and suppression system retained with the capsule and sized for residual fuel, hydraulic and ignition hazards that may remain after partial or failed separation. Independence from airframe-side systems is required where the safety assessment demands it. Status: [TARGET/REQ].

7.8 Parachute recovery system

Distributed multi-canopy architecture. A planning reference of eight canopies is retained for study purposes only. Final canopy count, type, total area, reefing and reliability allocation shall be determined by sizing and reliability analysis. Deployment is authorised only after capsule-clear logic is satisfied. Status: [REQ/TBD].

7.9 Summary of capsule protection layers (planning)

Layer	Function	Status
L1 – Aerial separation	Controlled jettison of defined mass/fire hazards before capsule landing	[REQ]
L2 – Parachute recovery	Controlled descent rate after capsule-clear (planning ref. 8 canopies)	[REQ/TBD]
L3 – Four-point shock-isolated attachments	Normal-flight load path + energy management + clean release	[TARGET/REQ]
L4 – CFRP plus-shaped roll-cage	Primary survival volume and impact load path	[TARGET/REQ]
L5 – Floor mechanical attenuation	Progressive energy absorption at touchdown / residual impact	[REQ/TBD]
L6 – Seat dual mechanical dampers	Additional progressive isolation between seat and roll-cage	[TARGET]
L7 – Thick energy-absorbing seats (≥ 50 cm)	Occupant support surface and further energy absorption	[TARGET/REQ]
L8 – 12 distributed airbags	Multi-direction residual impact protection + post-parachute near-ground	[TARGET]
L9 – Multi-layer fire suppression	Residual fire/ignition control after incomplete separation	[TARGET/REQ]
L10 – Post-event recovery aids	Location, life support and rescue access provisions	[REQ]

These layers are distinguished for design and analysis. They are not assumed to be automatically independent. Common-cause, cascading and partial-failure cases remain mandatory. No layer receives survival credit until the corresponding evidence is accepted.

8. RAPID SEPARATION SEQUENCE (DEVELOPMENT OBJECTIVE)

The ≤ 6 s sequence remains a development objective only. Each phase requires validated logic, state sensing and acceptable dynamics before the next phase is authorised.

Phase	Development objective	Engineering condition
1. Command validation	0 – 0.4 s	Validated command, inhibit checks and safe release-state establishment
2. Primary isolation / release	0.4 – 2.2 s	Only releases proven safe for the current dynamic state
3. Secondary release	2.2 – 3.8 s	Release additional designated items only if trajectory / loads remain acceptable
4. Capsule release	3.8 – 5.4 s	Release remaining capsule attachments and verify state
5. Capsule-clear decision	5.4 – 6.0 s	Authorise parachute deployment only if clearance logic is satisfied

9. SAFETY PHILOSOPHY AND RESIDUAL PROTECTION

Safety functions, independent initiation paths, physical separation functions and post-event protection are distinguished rather than treated as automatically independent layers. Candidate jettison items are not pre-credited as safely releasable. Partial and asymmetric separation, common-cause failures and downstream state authorisation are first-class design cases.

The multi-layer capsule protection chain (roll-cage + floor attenuation + seat dampers + thick seats + 12 airbags + fire suppression) is intended to provide residual occupant protection when aerial separation or parachute performance is incomplete. No quantitative statement of occupant G-load reduction or survival probability under extreme impact conditions (including hypothetical high-g events) is made. Such performance requires validated analysis, anthropomorphic test device simulation and physical test evidence before any claim is permitted.

10. PRINCIPAL OPEN CLOSURE ITEMS

- Bottom-up capsule mass, CG and inertia closure at the updated planning mass.
- Structural substantiation of the plus-shaped CFRP roll-cage, nodes and external surface integrity under normal, separation and impact load cases.
- Four-point attachment stiffness, load sharing, release dynamics, failed-attachment and unintended-release cases.
- Combined attenuation chain: floor mechanical elements + seat dual dampers + seat crush + airbag interaction.
- Airbag inflation timing, coverage, reliability, non-deployment and early-deployment cases.
- Occupant injury criteria compliance for the full seat / damper / airbag / structure chain.
- Power-required versus power-available model consistent with the 1.6 thrust-to-weight study target (or a revised target).
- Post-separation 6-DOF dynamics, component trajectories and parachute-clearance analysis.
- Quantitative unintended-jettison probability allocation and common-cause analysis.
- Fire-suppression effectiveness for residual fuel / ignition scenarios after incomplete separation.
- Door / hatch count, location and emergency-egress validation.
- Exact capsule external geometry (near-square / box ratios) versus internal volume and structural efficiency.

11. PRINCIPAL INTERFACES (MINIMUM CONTROLLED CONTENT)

Interface	Minimum controlled content
Capsule $\leftrightarrow$ airframe	Static / dynamic loads, CG, attachment stiffness, shock-isolation law, release sequence, clearance
Release mechanism $\leftrightarrow$ structure	Retention loads, actuation energy, fatigue, release timing, failure modes
Fuel $\leftrightarrow$ separation boundary	Isolation, residual pressure, leakage, fire, line-break and ignition control

Interface	Minimum controlled content
Electrical / power ↔ emergency system	Independent power, isolation, inhibits, fault containment and EMC
Crew ↔ emergency system	Controls, indication, arming, confirmation, workload and inadvertent-actuation protection
Capsule ↔ parachute	Deployment command, suspension loads, clearance, failure states and stability
Capsule ↔ landing attenuation	Load transfer, stroke, damping law, failed-leg cases and inspection
Seat ↔ roll-cage	Damper load-stroke law, attachment strength, bottoming and inspection
Airbag ↔ structure / seats	Inflation volume, pressure, timing, coverage and interaction loads

12. BASELINE ACCEPTANCE STATEMENT

Engineering disposition: LSH 00 Revision 3.3 is suitable as the root system-level preliminary baseline for controlled decomposition into a large engineering document tree and for further family-variant development under a common Safety Core. It is intentionally not a claim that the concept is already feasible, certifiable or safe in operation. All new geometric, mass, rotor, power, airbag, damper, door and G-load related figures remain planning values pending analysis, simulation and objective evidence. No survival probability under extreme impact conditions is asserted.

END OF LSH 00 — LIGHTWEIGHT SAFE HELICOPTER

Generation 1 | Revision 3.3 | Multi-Layer Capsule Protection | Preliminary Controlled System Baseline

Founder: Amirbahador Atai | Not a certification approval | No unverified performance represented as demonstrated